1. Siendo $f(x,y) = x^3 - 6xy + 3xy^2$ con $(x,y) \in \Re^2$, **grafique** el conjunto de puntos H^+ del plano xy para el cual el determinante hessiano $H(x,y)$ de f resulta positivo y **analice** en cuáles de dichos puntos la función f produce extremo local, **clasificándolo** y **calculando** su valor.
Nota: interprétese "determinante hessiano" o bien "determinante de la matriz hessiana".

$$f'_x = 3x^2 - 6y + 3y^2$$

$$f'_y = -6x + 6xy$$

$$f''_{xx} = 6x$$

$$f''_{xy} = -6 + 6y$$

$$f''_{yx} = -6 + 6y$$

$$F''_{yy} = 6x$$

$$Hf = \begin{bmatrix} 6x & -6+6y \\ -6+6y & 6x \end{bmatrix}$$

$$Det(Hf) = 36x^2 - (-6+6y)^2$$

$$36x^2 - (-6+6y)^2 > 0$$

$$36x^2 > (-6+6y)^2$$

$$\cancel{36}x^2 > \cancel{36}(-1+y)^2$$

$$x^2 > (y-1)^2$$

$$\sqrt{x^2} = \sqrt{(y-1)^2}$$

$$|x| = |y-1|$$

$$\frac{(x-x_0)^2}{a^2} + \frac{(y-y_0)^2}{b^2} = 1$$

$$y - y_0 = a(x-x_0)^2$$

$$\frac{(x-x_0)^2}{a^2} - \frac{(y-y_0)^2}{b^2} = 1$$

$$|x| = |y-1|$$

$x > 0 \quad y-1>0 \longrightarrow x = y-1$
$\quad y>1 \qquad y = x+1$

$x<0 \quad y>1 \longrightarrow -x = y-1$
$\qquad y = -x+1$

$x>0 \quad y<1$
$x = -y+1$
$y = -x+1$

$x<0 \quad y<1$
$y = x+1$

$|x| = |y-1|$

$x>0 \quad y-1>0 \quad y>1 \longrightarrow x = y-1$

$y = x+1$

$x<0 \quad y>1 \longrightarrow -x = y-1$

$y = -x+1$

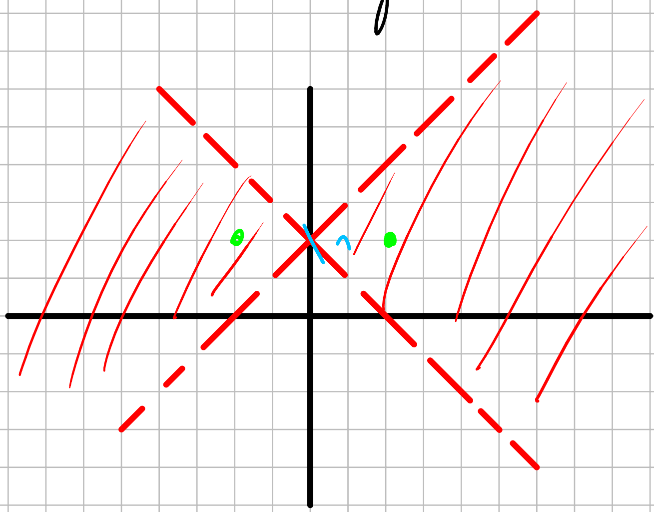
$x>0 \quad y<1$

$x = -y+1$

$y = -x+1$

$x<0 \quad y<1$

$y = x+1$



Abierto ✓
Cerrado ✗
Acotado ✗
Compacto ✗
Conexo ✗

$$\nabla f = (3x^2 - 6y + 3y^2, -6x + 6xy) = (0,0)$$

$3x^2 - 6y + 3y^2 = 0 \quad \wedge \quad -6x + 6xy = 0$

$x(-6+6y) = 0$

$x = 0$

ó $y = 1$

$-6y + 3y^2 = 0$

$\underbrace{3y(-2+y) = 0}$

$y = 0$

ó $y = 2$

$3x^2 - 6 + 3 = 0$

$3x^2 = 3$

$x^2 = 1$

$x = \pm 1$

Ptos crit:

$(0,0)$

$(0,2)$

$(1,1)$ ✓

$(-1,1)$ ✓

$(1,1)$

$(-1,1)$

$$Hf = \begin{bmatrix} 6x & -6+6y \\ -6+6y & 6x \end{bmatrix}$$

$$HF(1,1) = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} \quad \begin{matrix} H_1 = 6 \\ H_2 = 36 \end{matrix} \Big\} \text{MIN} \qquad f(1,1) = -2$$

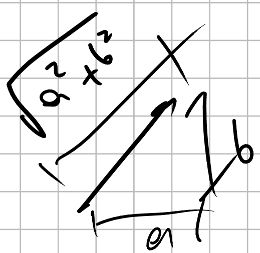
$$HF(-1,1) = \begin{bmatrix} -6 & 0 \\ 0 & -6 \end{bmatrix} \quad \begin{matrix} H_1 = -6 \\ H_2 = 36 \end{matrix} \Big\} \text{Max.} \qquad f(-1,1) = 2$$

---

2. Dada $f(x,y) = \begin{cases} \dfrac{\operatorname{sen}(2x^3 + xy)}{x^2+y^2} & \text{si } (x,y) \neq (0,0) \\ 0 & \text{si } (x,y) = (0,0) \end{cases}$ **Analice** si f admite derivada parcial de 1º orden respecto de la variable x para todo $(x,y) \in \Re^2$.

$$f'_x = \frac{\cdots\cdots}{(x^2+y^2)^2} \quad \neq (0,0)$$

$\longrightarrow$ admite derivada $\forall (x,y) \neq (0,0)$



$\vec{r} = (a,b)$

$\sqrt{a^2+b^2} = 1$

$a^2+b^2 = 1$

$$\lim_{h\to 0} \frac{f(A+h\vec{r}) - f(A)}{h}$$

$A = (0,0)$

$\vec{r} = (1,0) = \hat{i}$

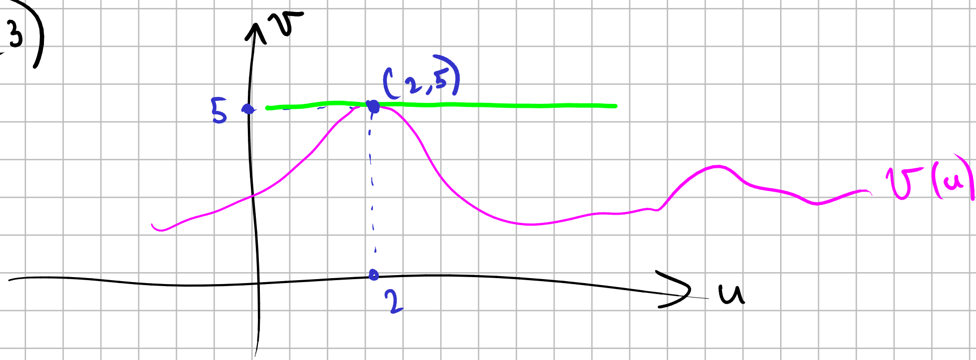
$$\lim_{h\to 0} \frac{f((0,0)+h(1,0)) - \cancel{f(0,0)}^{\,0}}{h}$$

$$\lim_{h\to 0} \frac{f(h,0)}{h} = \lim_{h\to 0} \frac{\operatorname{sen}(2h^3)\cdot 2}{2h^3} = 2$$

4. Dada $h(x,y)=f(\overbrace{x^2-y^2}^{g(x,y)})$ con $f\in C^1(\Re)$, **calcule** una aproximación lineal para $h(2.02,0.99)$ sabiendo que la curva de ecuación $v=u^2+f(u^2-1)$ del plano uv tiene recta tangente horizontal (paralela al eje u) en el punto $(2,5)$.

$$h(x,y)\approx h(2,1)+h'_x(2,1).(x-2)+h'_y(2,1).(y-1)$$

$$h(2,1)=f(3)$$



$$5=4+f(3)$$

$$\boxed{f(3)=1}$$

$$h(2,1)=1$$

$$h(x,y)=f(g(x,y))$$

$$Dh(2,1)=Df(g(2,1)).Dg(2,1)$$

$$Dh(2,1)=\underbrace{Df(3)}_{f'(3)}.Dg(2,1)$$

$$g(x,y)=x^2-y^2$$

$$Dg=[2x \quad -2y]$$

$$Dg(2,1)=[4 \quad -2]$$

$$v=u^2+f(u^2-1)$$

$$v'=2u+f'(u^2-1).2u$$

$$v'(2)=4+f'(3).4=\underbrace{0}_{\text{Recta tg horiz.}}$$

$$f'(3)=-1$$

$$Dh(2,1) = Df(3) \cdot Dg(2,1)$$

$$Dh(2,1) = -1 \cdot \begin{bmatrix} 4 & -2 \end{bmatrix}$$

$$Dh(2,1) = \begin{bmatrix} -4 & 2 \end{bmatrix}$$

$$h(x,y) \approx 1 - 4(x-2) + 2(y-1)$$

$$h(2.02, 0.99) \approx 1 - 4 \cdot (2.02 - 2) + 2 \cdot (0.99 - 1) = 0{,}9$$

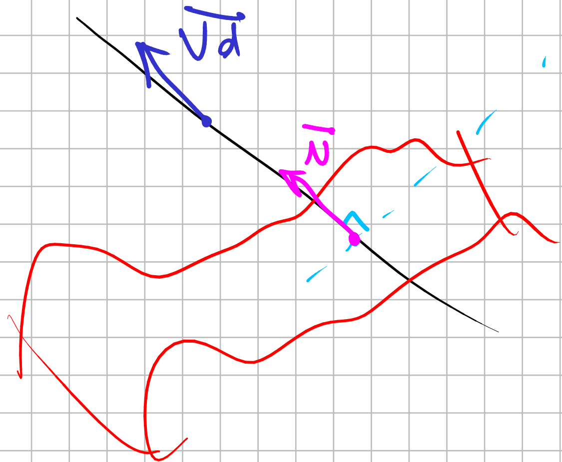
$$h(2.02, 0.99) \approx 0{,}9$$

---

5. **Halle** el punto (x_0, y_0, z_0) donde la recta definida por la intersección de los planos de ecuaciones $x - y + z = 7$ y $2y - x + 3z = 18$ es normal a la superficie de ecuación $z = x + yx^2$.

$$\begin{array}{r} x - y + z = 7 \\ + \quad 2y - x + 3z = 18 \\ \hline y + 4z = 25 \end{array}$$

$$y = 25 - 4z$$



$$x = 7 + y - z$$

$$x = 7 + 25 - 4z - z$$

$$x = 32 - 5z$$

$$(x,y,z) = (32 - 5z,\ 25 - 4z,\ z)$$

$$= (-5z, -4z, z) + (32, 25, 0)$$

$$= z(-5, -4, 1) + (32, 25, 0)$$

$$\underbrace{x + yx^2 - z}_{F} = 0$$

$$\nabla F \perp F = k$$

$$\nabla F = (1 + 2xy,\ x^2,\ -1) = \underbrace{k}_{-1}(-5, -4, 1)$$

$$(1 + 2xy,\ x^2,\ -1) = (5, 4, -1)$$

$$(1+2xy,\ x^2,\ -1) = (5,4,-1)$$

$$1+2xy=5$$

$$\underline{x^2=4}$$

$$\left.\begin{array}{l} x=2 \\ x=-2 \end{array}\right] \quad \begin{array}{l} \to 1+4y=5 \quad y=1 \\ \to 1-4y=5 \quad y=-1 \end{array}$$

$A:(2,1,z_A)$

$B:(-2,-1,z_B)$

$$z = x + yx^2 .$$

$z_A = 6$

$z_B = -6$

$\boxed{A=(2,1,6)}$

$\underline{B=(-2,-1,-6)}$

No pertenece a la recta

$$(32-5z,\ 25-4z,\ z)$$

$\hookrightarrow z=6 \ = (2,1,6)$ ✓

$\hookrightarrow z=-6 = (6\ 2, \ldots\ \ldots)$ ✗.

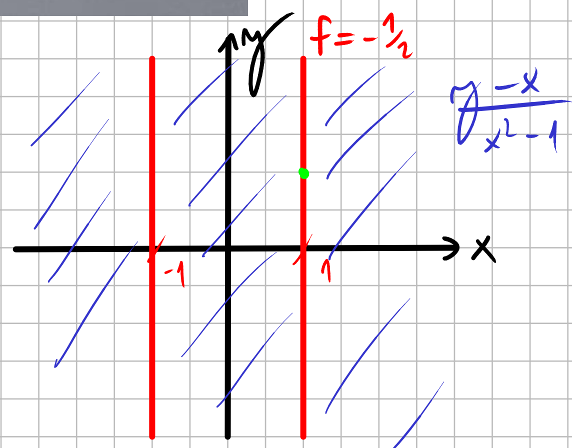
Ejercicio 1 Consideremos la función $f:\mathbb{R}^2\to\mathbb{R}$ dada por $f(x,y)=\begin{cases}\dfrac{y-x}{x^2-1} & \text{si } |x|\neq 1\\ -\frac{1}{2} & \text{si } |x|=1\end{cases}$.

1. ¿Es f continua en $(1,1)$? ¿Es f diferenciable en $(1,1)$?

$$\lim_{(x,y)\to(1,1)} f(x,y)$$

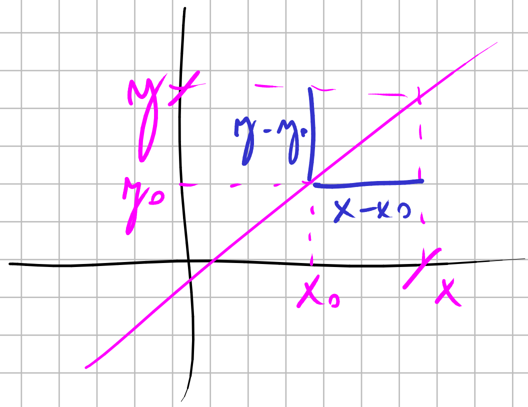
$$\lim_{(x,y)\to(1,1)} -\frac{1}{2} = -\frac{1}{2}$$

$$\lim_{(x,y)\to(1,1)} \frac{y-x}{x^2-1}$$



$$y - y_0 = m(x - x_0)$$

$y - y_0 = m(x - x_0)$



$m = \dfrac{y - y_0}{x - x_0}$

$y - 1 = m(x-1)$

$y = 1 + m(x-1)$

$$\lim_{x\to 1} \frac{1+m(x-1)-x}{x^2-1}$$

$$L'H \lim_{x\to 1} \frac{m-1}{2x} = \frac{m-1}{2} \to \nexists \lim \Rightarrow \text{No es cont.}$$

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L \Rightarrow \lim_{(x,y)\underset{C}{\to}(x_0,y_0)} f(x,y) = L$$

$\forall C / (x_0,y_0) \in C$

$A \Rightarrow B$

$\neg B \Rightarrow \neg A$

No es cont $\Rightarrow$ No es dif.

$$f'(A, \check{r}) = \nabla F(A) . \underbrace{\check{r}}_{(a,b)}$$

Ejercicio 2. Asumamos que $f : \mathbb{R}^2 \to \mathbb{R}$ es una función de tipo C^2 tal que

$$\nabla f(3,-2) \perp (1,-1) \qquad \text{y} \qquad f_{xx}(3,-2) = f_{yy}(3,-2) + 3.$$

Calcular $p(4,-1)$ siendo p el polinomio de Taylor de primer orden en $(3,-2)$ de la función

$$g(x,y) = f_x(x,y) - f_y(x,y).$$

$f''_{xx} - f''_{yy} = 3$

$$p(x,y) = g(x_0,y_0) + g'_x(x_0,y_0)(x-x_0) + g'_y(x_0,y_0)(y-y_0)$$

$$p(x,y) = g(3,-2) + g'_x(3,-2)(x-3) + g'_y(3,-2)(y+2)$$

$$g'_x = f''_{xx} - f''_{yx}$$

$+$

$$g'_y = f''_{xy} - f''_{yy}$$

$$g'_x + g'_y = f''_{xx} - f''_{yy} = 3$$

$$p(4,-1) = \overset{0}{\cancel{g(3,-2)}} + \underbrace{g'_x(3,-2) + g'_y(3,-2)}_{3}$$

$$p(4,-1) = 3$$

$$g(3,-2) = f'_x(3,-2) - f'_y(3,-2) = 0$$

$$\nabla f(3,-2) = (f'_x(3,-2), f'_y(3,-2)) \perp (1,-1)$$

$$f'_x(3,-2) - f'_y(3,-2) = 0$$

3. (2.5 pts.) Dada la función $f(x,y) = a(x^2 - 2x) + y^2 + (a^2 + 3a)y$ con $a \in \mathbb{R}$.

(a) (1 pt.) Estudiar, para cada valor de a, si el punto $(2,0)$ es un extremo de f (aclarar el tipo), un punto silla o ninguno de los anteriores.

(b) ~~(1.5 pts.) Para $a = -3$, hallar los extremos globales de f en el conjunto~~

$$\cancel{A = \{(x,y) \in \mathbb{R}^2 : x^2 + 9y^2 \leq 9\}.}$$

$$\nabla f = (a(2x-2),\ 2y + a^2 + 3a)$$

$$\nabla f(2,0) = \underbrace{(2a,\ a^2+3a)}_{\to a=0} = (0,0)$$

$$f(x,y) = y^2 \quad (a=0) \to y=0$$

MIN. $(2,0)$

$$f''_{xx} = 2a = 0$$

$$f''_{xy} = 0$$

$$f''_{yx} = 0$$

$$f''_{yy} = 2$$

$$Hf(2,0) = \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}}_{}$$

criterio no decide ☺

22. Dada $y = f(x,z)$ definida implícitamente por $\overbrace{x\,y + x\,z + \ln(1+y\,z) - 4}^{F} = 0$ en un entorno de $(x_0, z_0) = (2,0)$, **calcule** aproximadamente $f(1.98, 0.03)$ mediante Taylor de 1^0 orden.

$$f(x,z) \approx f(2,0) + f'_x(2,0)(x-2) + f'_z(2,0).z$$

$$2y_0 - 4 = 0$$
$$y_0 = 2 = f(2,0)$$

I) $F(2,2,0) = 0$

II) $\nabla F = \left(y+z,\ x + \frac{z}{1+yz},\ x + \frac{y}{1+yz}\right)$ cont. en $E(2,2,0)$

III) $F'_y(2,2,0) = 2 \neq 0.$

$$f'_x(2,0) = -\frac{F'_x(2,2,0)}{F'_y(2,2,0)} = -\frac{2}{2} = -1$$

$$f'_z(2,0) = -\frac{F'_z(2,2,0)}{F'_y(2,2,0)} = -\frac{4}{2} = -2$$

$$f(x,z) \approx 2 - 1(x-2) - 2z$$

$$f(1.98, 0.03) \approx 2 - 1(1.98-2) - 2.0{,}03 = 1{,}96$$

$$f(1.98, 0.03) \approx 1{,}96$$

27. Sea $f : \mathbb{R}^3 \to \mathbb{R}$ con $f \in C^1(\mathbb{R}^3)$. Sabiendo que $\vec{X} = (u\,,\, v^2\,,\, v)$ con $(u,v) \in \mathbb{R}^2$ es una ecuación de la superficie de nivel 1 de f, que para $A = (2,1,1)$ resulta $f(A) = 1$ y que la derivada direccional $f'(A\,,\,\check{r}) = 3$ para $\check{r} = (1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$, **calcule** el valor de la mínima derivada direccional de f en el punto A.

$$f'_{MIN}(A) = -\|\nabla f(A)\|$$

$$\vec{a}\cdot\vec{b} = \|\vec{a}\|\|\vec{b}\|.\cos\theta$$

$$f'(A,\check{r}) = \nabla f(A).\check{r} = \|\nabla f(A)\|.\overbrace{\|\check{r}\|}^{1}.\cos\theta$$

f dif

$$= \|\nabla f(A)\|.\cos\theta$$

$$\underbrace{-\|\nabla f(A)\|}_{f'_{MIN}(A)} \leq \|\nabla f(A)\|.\cos\theta \leq \underbrace{\|\nabla f(A)\|}_{f'_{MAX}(A)}$$

$$\check{r}_{MIN} = -\frac{\nabla f(A)}{\|\nabla f(A)\|}$$

$$\check{r}_{MAX} = \frac{\nabla f(A)}{\|\nabla f(A)\|}$$

$$\check{r}_{NULOS} \perp \nabla f(A)$$

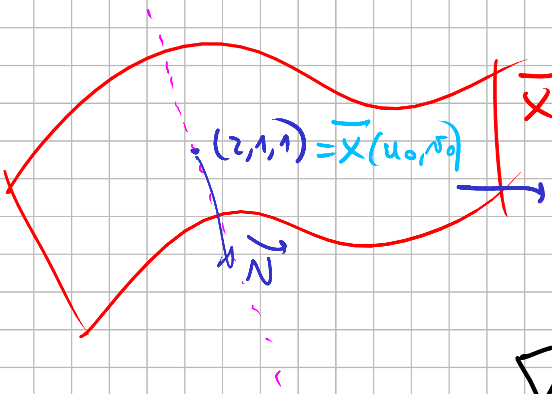
$$\nabla f(A) = \underbrace{(f'_x(A), f'_y(A), f'_z(A))}_{(a,b,c)}$$

$$f'_{MIN}(A) = -\|\nabla f(A)\|$$

$$f'\left(A, \left(\tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}\right)\right) = \nabla f(A).\left(\tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}, \tfrac{1}{\sqrt{3}}\right) = 3$$

$$\frac{a}{\sqrt{3}} + \frac{b}{\sqrt{3}} + \frac{c}{\sqrt{3}} = 3$$

$$a+b+c = \sqrt{3}.3$$



$\vec{x} = (u, v^2, v)$

$\to f = 1$

$\nabla f \perp f=1$

$\nabla f(A) \perp f=1$ en A.

$$\vec{x}'_u = (1,0,0)$$
$$\times\ \vec{x}'_v = (0, 2v, 1)$$
$$\overline{(0,-1,2v)}$$

$$\vec{x}(u_0,v_0) = (u_0, v_0^2, v_0) = (2,1,1)$$

$u_0 = 2$

$v_0 = 1$

$$\vec{N}(A) = (0,-1,2)$$

$$\nabla f(A) = K(0,-1,2)$$
$$= (0, -K, 2K)$$

$$-K + 2K = \sqrt{3}\cdot 3$$
$$K = 3\sqrt{3}$$

$$\nabla f(A) = (0, -3\sqrt{3}, 6\sqrt{3})$$

$$f'_{MIN}(A) = -\|(0,-3\sqrt{3}, 6\sqrt{3})\| = -\sqrt{27+108}$$
$$= -\sqrt{135}$$

21. Sea $f : \mathbb{R}^2 \to \mathbb{R}$ definida por:

$$f(x,y) = \begin{cases} \dfrac{2\,x\,y}{3\,x^2 + y^2} & \text{si} \quad (x,y) \neq (0,0) \\ 0 & \text{si} \quad (x,y) = (0,0) \end{cases}$$

Analice la continuidad de f en los distintos puntos de $\mathbb{R}^2$.

Fuera de (0,0) es una división de polinomios con denominador no nulo => continua.

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} \frac{2xy}{3x^2+y^2}$$

$$y = mx \qquad \lim_{x\to 0} \frac{2x.mx}{3x^2+m^2x^2} = \lim_{x\to 0} \frac{\cancel{x^2}}{\cancel{x^2}} \cdot \frac{2m}{3+m^2} = \frac{2m}{3+m^2}$$

$\nexists$

$$f(x,y) = \begin{cases} \dfrac{2\,x\,y^4}{3\,x^2 + y^2} & \text{si} \quad (x,y) \neq (0,0) \\ 0 & \text{si} \quad (x,y) = (0,0) \end{cases}$$

$$\lim_{(x,y)\to(0,0)} \frac{2x\,y^2 . y^2}{3x^2+y^2} = 0$$

acot

acot.

$$\frac{a}{a+b} \qquad a \geq 0 \quad b \geq 0$$

$$\frac{a}{\sqrt{a^2+b^2}} \quad \text{acot.}$$

$$0 \leq y^2 \leq y^2 + 3x^2$$

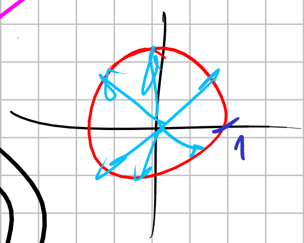
$$0 \leq \frac{y^2}{y^2+3x^2} \leq 1$$

$$0 \leq \frac{|y|}{\sqrt{y^2+3x^2}} \leq 1 \qquad -1 \leq \frac{y}{\sqrt{y^2+3x^2}} \leq 1$$

$$f \text{ dif} \Rightarrow \begin{cases} f \text{ cont} \\ f \text{ deriv} \\ f'(A,\breve{r}) = \nabla f(A) \cdot \breve{r} \quad \forall \breve{r}. \end{cases}$$

$$f'(A,\breve{r}) = \frac{a^2 b}{\underbrace{2-a}_{\text{deriv } \forall (a,b)}}$$

$$a^2+b^2=1$$



$$\nabla f(A) = \left(f'_x(A), f'_y(A)\right)$$

$$f'_x(A) = f'(A,(1,0)) = 0 \qquad \nabla f(A) = (0,0)$$

$$f'_y(A) = f'(A,(0,1)) = 0$$

$$\nabla f(A) \cdot \breve{r}$$

$$= (0,0)\cdot(a,b) = 0 \neq \frac{a^2 b}{2-a}$$

↳ No es dif.

$$\frac{ab}{\left(a-\frac{\sqrt{3}}{2}\right)^2 + \left(b-\frac{\sqrt{3}}{2}\right)^2}$$

$$(a,b) = \left(\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right) \qquad a^2+b^2=1$$

$$\frac{3}{4}+\frac{3}{4} \neq 1.$$

$$\frac{a \cdot b}{\frac{1}{2}-a} \qquad a \neq \frac{1}{2}$$

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$
$$\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right) \rightarrow \nexists \text{ deriv.}$$

$$a^2+b^2=1$$
$$\frac{1}{4}+b^2=1$$
$$b^2=\frac{3}{4}$$
$$b=\pm\frac{\sqrt{3}}{2}$$